Finesse × Citadel

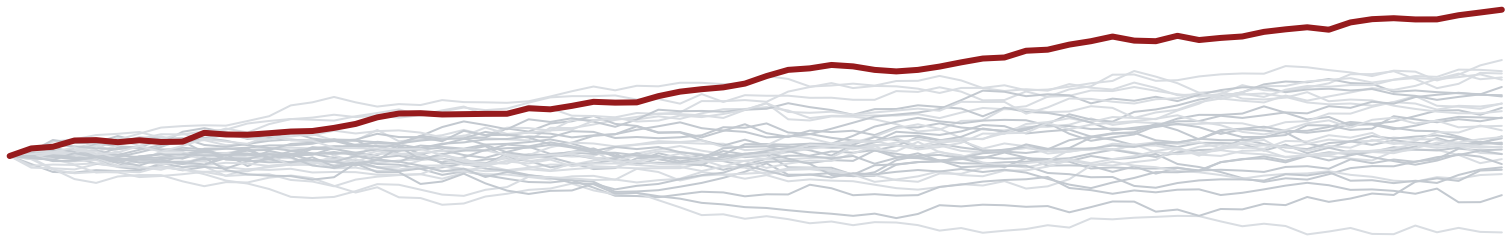
Portfolio Construction Challenge · Round 2 · Jury round

12-1 Momentum Strategy

Buy the 10 strongest stocks of the Nifty 100 and Midcap 100, equal weight, reset every month

Tested against 10,000 random portfolios · Jan 2021 to Dec 2025

Illustration: random portfolios (gray) against our rule (red). Not real data.



Rationale · Robustness · Risk management · Transparency · Portfolio logic

TEAM

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Evidence base | 63 runs, 10,000 random portfolios and 125 tests back the result

10,000

random 10-stock portfolios

Same 200 stocks, dates, costs and engine. Their spread is the luck benchmark.

63

runs, including 58 alternatives

Each slate's predictions were written down before it ran. Losers are published.

1,786

trading days of cleaned prices

Jun 2019 to Aug 2026. 4 fake holiday bars and 1 stale day removed.

79

stock splits checked

3 false crashes fixed. Demergers are exited, never adjusted with an invented ratio.

43

index changes rebuilt

From 27 NSE press releases found in 976 PDFs, to measure the stock-list bias.

125

automated tests

Cover causality, accounting, cleaning and selection. The headline reproduces in about 6 minutes.

Every number is reproducible from the team's repository, including the runs that failed

Sources: docs/PROJECT.md §11 trial ledger, data/reports/data_quality.md, output/report/point_in_time, tests/

Recommendation summary

A simple momentum rule beat luck by a wide margin, and we measured what flatters it

1	**Rationale**	Investors under-react to news, so trends persist for 3–12 months. The rule needs only prices: rank 12-1 momentum, hold the top 10 at 10% each, reset monthly.
2	**Robustness**	0 of 10,000 random portfolios beat it (+11.1σ) and none of 58 alternatives did better. It ranks at the 96th percentile on return per unit of risk.
3	**Risk and transparency**	Drawdown reached −32.4% with no stop-loss, and the gain depends on one regime. On each date's real index lists, +1,076% falls to +414%.

₹10.76 Cr net profit on ₹1 Cr | CAGR 63.8% | Sharpe 2.42 | after 0.1% costs

Strategy | One ranking, ten stocks, equal weight, reset every month

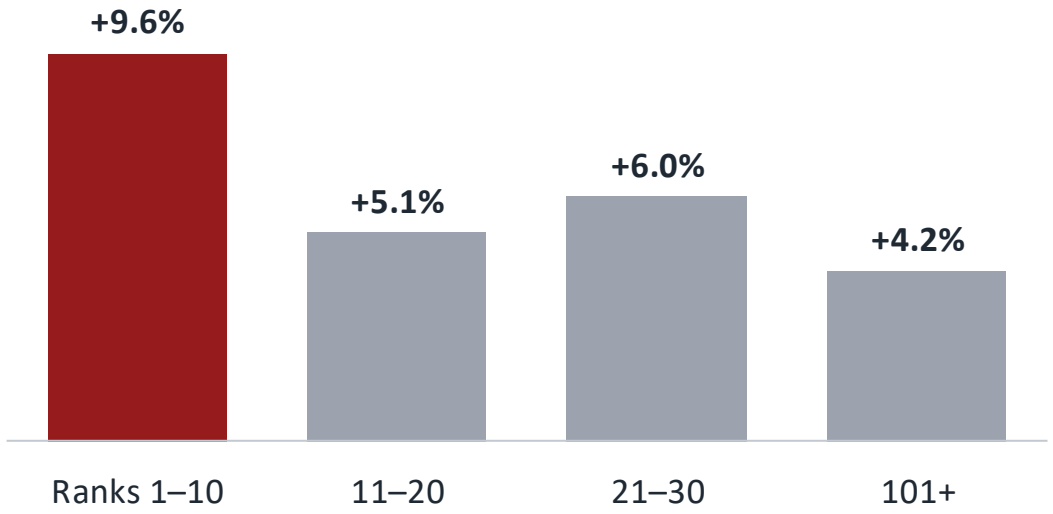
$$MOM_{i,t} = \frac{P_{i,t-21}}{P_{i,t-252}} - 1$$

P = daily close adjusted for splits. The latest month (21 trading days) is skipped.

How it works

1. **Universe:** Today's Nifty 100 + Midcap 100, 200 names. A stock needs a full year of prices: 174–192 qualify.
2. **Signal:** Return from 252 to 21 trading days ago, using closes through yesterday.
3. **Select:** One ranking across both indices. Top 10, ties go to the current holder. No optimiser.
4. **Hold:** 10% each, reset on the first trading day of every month. 60 rebalances, 0.1% on every trade.

Next-quarter return by momentum rank



No look-ahead: signals use closes through yesterday and trades fill at today's open. Scrambling every price from the rebalance date onward leaves the same 10 picks, and scrambling the lookback window changes them.

Source: team numbers report and docs/NOTES.md N7. Rank chart: point-in-time lists, 20 quarterly dates, no error bars; ranks 21–30 beat 11–20, so the pattern is not perfectly monotonic.

Why it works | Investors react slowly to news, so past winners keep winning for 3–12 months

What the research shows

Jegadeesh & Titman (1993)	Stocks that won over the past 3–12 months kept beating past losers over the next 3–12 months.
Jegadeesh (1990)	Returns reverse over the most recent month, so the rule skips it.
Asness, Moskowitz & Pedersen (2013)	Momentum appears across countries and asset classes, not one market.
Daniel & Moskowitz (2016)	Momentum can crash after market falls and sharp rebounds.

Why the pattern persists

1. **Under-reaction**
   Investors adjust slowly to news, so prices drift in the same direction (Barberis, Shleifer & Vishny, 1998).

2. **Slow information spread**
   News reaches investors gradually, so trends build over months (Hong & Stein, 1999).

3. **Overconfidence**
   Investors over-trust their own signals, which pushes prices further along the trend (Daniel, Hirshleifer & Subrahmanyam, 1998).

How our 12-1 signal uses it: rank on the 11 months before last month, hold for 1 month



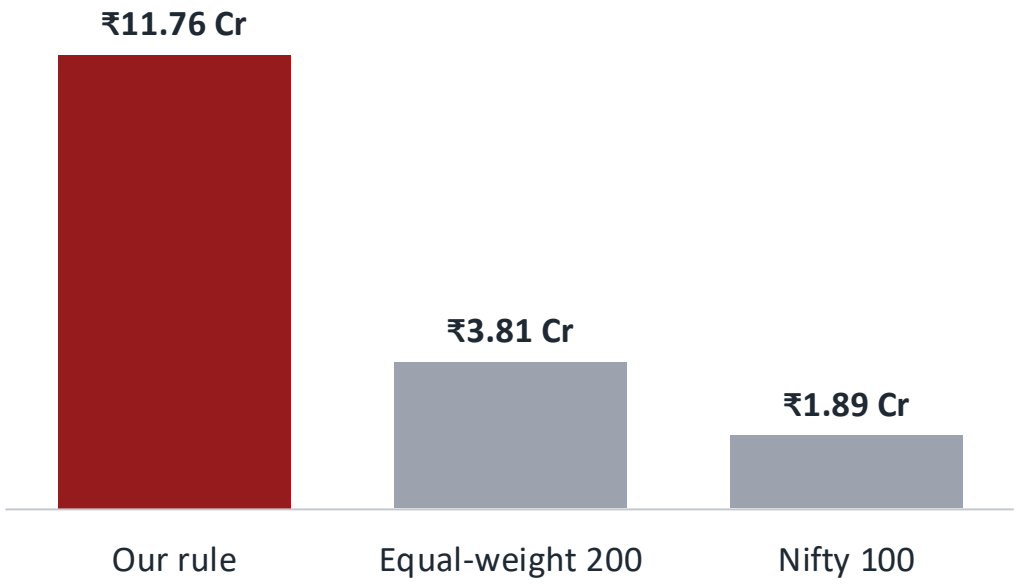
Formation window: 252 to 21 trading days back **Skipped month** **Hold month**

Our data: ranks 1–10 earned +9.56% next quarter, rank 101+ +4.19%.

Sources: J. Finance 48(1) 1993, 45(3) 1990, 68(3) 2013; J. Financial Economics 122(2) 2016; Barberis et al. 1998; Hong & Stein 1999.

Results | ₹1 Cr became ₹11.76 Cr and beat both benchmarks in every year

Value of ₹1 Cr by Dec 2025, ₹ Cr



Calendar-year return

Year	Our rule	Equal-weight	Nifty 100
2021	**+160.3%**	+50.3%	+25.0%
2022	**+22.2%**	+14.2%	+3.6%
2023	**+74.3%**	+52.0%	+20.0%
2024	**+86.4%**	+31.0%	+11.8%
2025	**+13.7%**	+10.6%	+9.0%

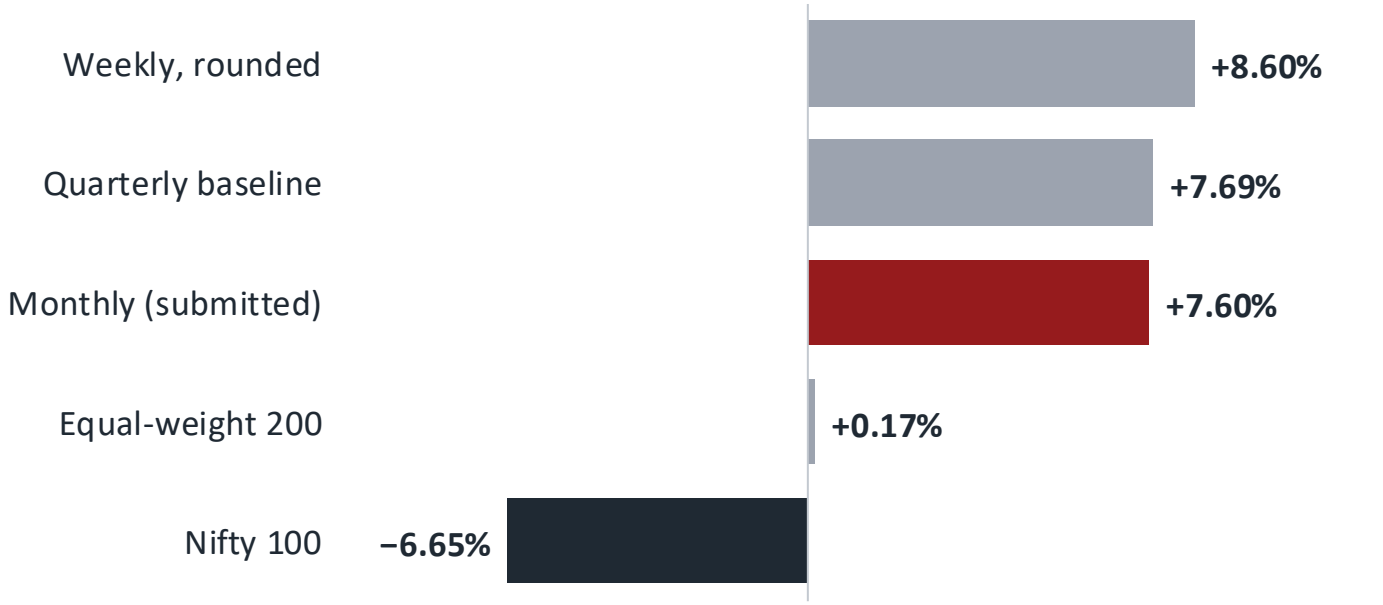
Against equal weight the rule is +796 pp; against the index, +987 pp.

63.8%	2.42	–32.4%	2.01	62.5%	7.36×
CAGR	Sharpe	Max drawdown	Gain-to-loss	Accuracy	Turnover a year

Source: numbers.md §1–§3. Equal-weight pays costs; Nifty 100 is cost-free. Price return only (dividends ≈ 9 pp). 2021 alone returned +160%.

Out-of-sample | Jan–Jun 2026: the submitted rule returned +7.60%, against +0.17% for equal weight and −6.65% for the Nifty 100

Return on a fresh ₹1 Cr, 1 Jan to 30 Jun 2026



88.9th
percentile of a fresh random band

+14.3 pp
ahead of the Nifty 100

How to read a six-month test

Rejection filter only. The 2026 window was used to drop candidates, never to choose one. Quarterly scored slightly higher, and we still submitted monthly because 2021–25 decides.

Small sample. Six months in one market regime is one observation.

Sensitive to data errors. One stale price row had lifted the quarterly result from +7.69% to +15.48%. Exiting VEDL one session before its demerger lifted a weekly variant from +1.8% to +8.6%.

Source: WALKTHROUGH.md §3, §6, §9. Each starts with ₹1 Cr. Only these 2026 results are published; other cells are in configurations.md.

Robustness | 0 of 10,000 random portfolios beat the rule; the median random book earned ₹2.36 Cr

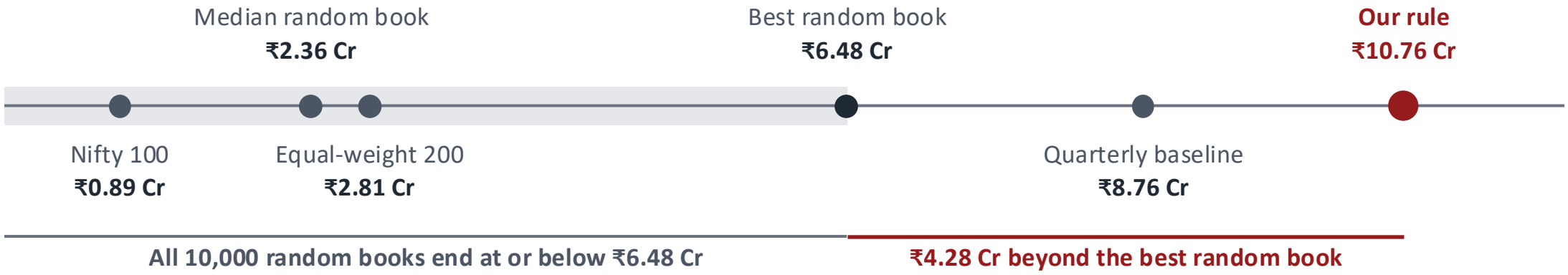
How the random test works

1 Draw
10,000 random 10-stock portfolios from the same 200 stocks.

2 Run
Same dates, 0.1% costs and engine. Only the stock picks differ.

3 Compare
Score our rule by its distance from the random spread, in σ and percentile.

Net profit on ₹1 Cr, 2021–25, ₹ Cr (drawn to scale)



+11.1σ
above the random spread

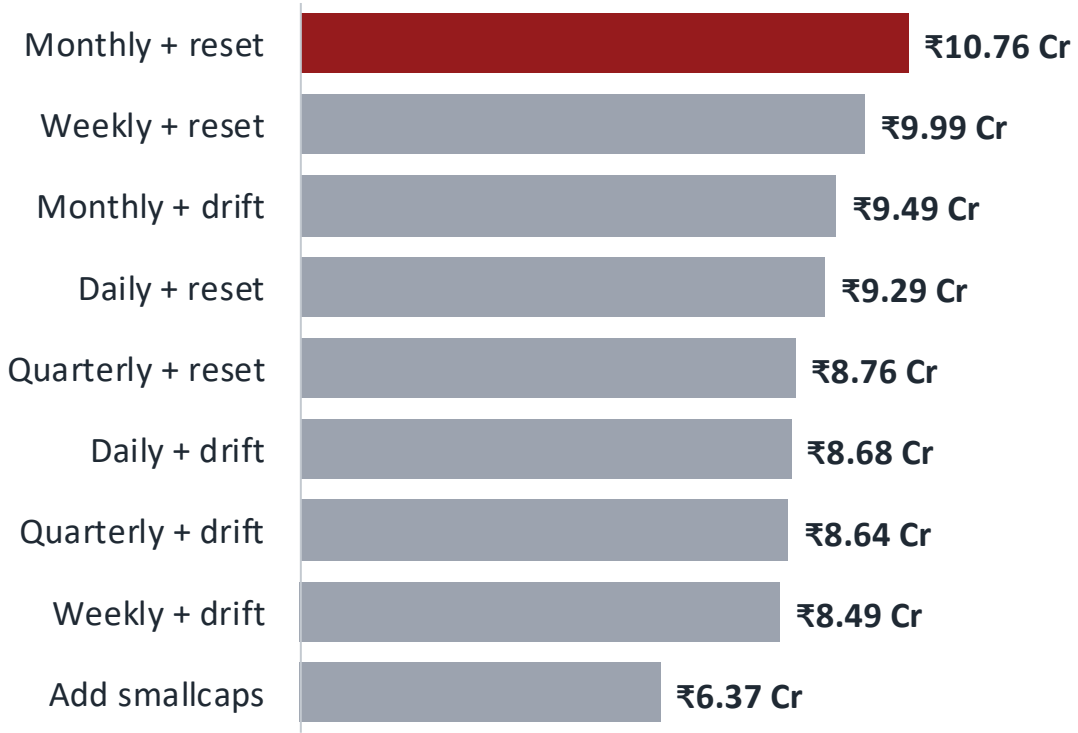
96th
percentile per unit of risk

Same test
used to score all 58 alternatives

Source: WALKTHROUGH.md §1, §5a. To scale; only the median, best draw and percentiles are published. Random books cannot remove the stock-list bias (slide 10).

Robustness | 58 alternatives were tried and none beat the simple monthly rule

Net profit by variant, ₹ Cr, 2021–25



Caveat: monthly was the best of 8 cells, so some selection effect remains.

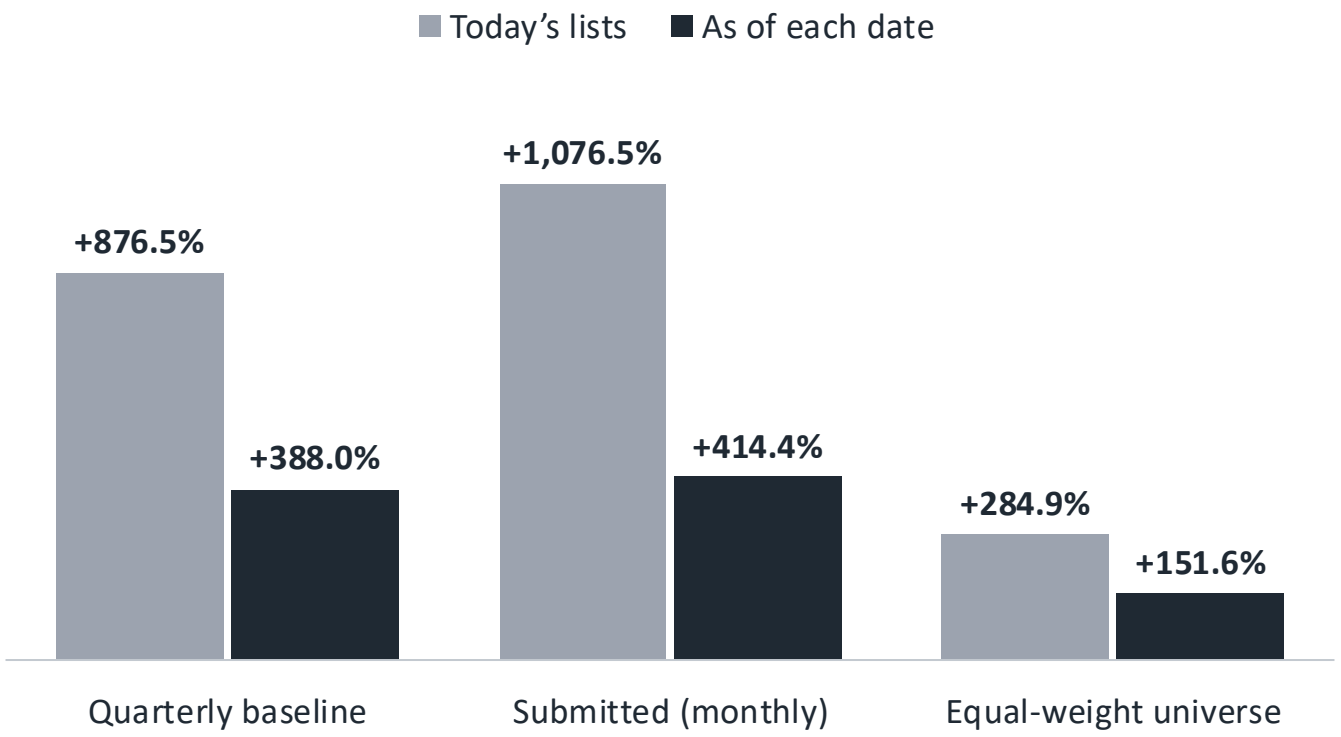
What each slate found

Slate	Result	Verdict
Frequency × weighting (8 runs)	Monthly + reset beats quarterly by ₹2.00 Cr (+2.32σ)	**Adopted**
Lookback × skip (6 runs)	252/21 days is the best of six	**Kept**
Add 99 smallcaps (2 runs)	Profit falls ₹4.39 Cr (−5.1σ)	**Rejected**
Add signals (4 runs)	All four lose by 5.15σ to 7.49σ	**Rejected**
Composite weights (42 runs)	0 of 42 beat the baseline	**Rejected**

Source: docs/PROJECT.md §5, §11. Smallcap bar = ₹10.76 Cr less ₹4.39 Cr. Per-variant drawdown and turnover are in output/report/configurations.md. 2026 window used only to reject.

Risk | Today's stock lists flatter the result: +1,076% falls to +414% on as-of-date lists

Five-year return by stock list, 2021–25



Where the bias comes from

Index inclusion

A stock is in today's Nifty 100 partly because it rose over 2021–25, so we pick winners from a list already filtered for winning.

Survivorship

Names that left the indices are absent, so the rule can never be caught holding one.

How we measured it

Rebuilt from 27 NSE releases (43 changes). Gap: 488 pp on the baseline, 662 pp on the submitted rule.

Over half the headline comes from the universe, not the rule. The edge over its own universe survives: +236 pp, beating 9,996 of 10,000 random books

Source: output/report/point_in_time/. Excluded: 6 former members with no usable prices (DHANI, GSPL, HDFC, ISEC, MINDTREE, PEL). The backtest itself uses no future prices.

Risk | A −32.4% drawdown, 10 concentrated stocks and one market regime, accepted to maximise profit

Risks we accepted on purpose

Drawdown. −32.4% with no stop-loss and always fully invested, from 20 Sep 2022 to 27 Feb 2023. Recovered 21 Jul 2023. The score is raw profit, so overlays would lower it.

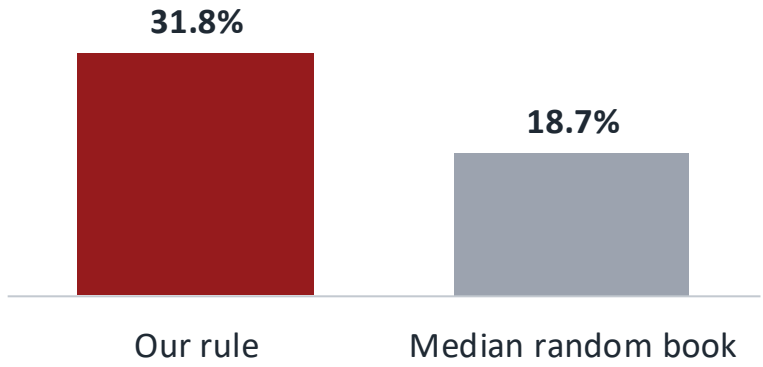
Concentration. 10 names at 10% each, in themes: Adani in 2021–22, PSU banks in 2023, rail, defence and power PSUs in 2024–25.

Momentum crash. Three Adani stocks cut the book 26.7% in Jan–Feb 2023; the rule kept and topped them up. We did not patch one known event: that would be overfitting.

One regime. 2021 alone returned +160%. The 2026 test covers six months: +7.6% against −6.7% for the Nifty 100.

Costs. A flat 0.1% per trade (₹20.6 L in total). STT, stamp duty, GST and slippage are not modelled separately.

Volatility, 2021–25



Risks we controlled

Look-ahead: causal signal, tested by price scrambling

Cash: costs reserved up front, never negative

Liquidity: 99% of fills under 1.82% of daily volume

Corporate actions: exit before demergers

Source: numbers.md §3, §5; WALKTHROUGH.md §8; holdings and NAV in output/sweep/monthly_reset. A live book would add position limits, sector caps and regime awareness.

Conclusion

1 | Why it earns returns

Investors under-react to news, so trends persist for 3–12 months. Ranks 1–10 earned +9.6% next quarter against +4.2% for rank 101+. The 12-1 rule has no fitted parameter.

2 | Why we trust it

0 of 10,000 random portfolios beat it (+11.1σ) and none of 58 alternatives did better. It sits at the 96th percentile per unit of risk. 125 tests back the code.

3 | What limits it

A −32.4% drawdown with no stop-loss, 10 concentrated names and one regime. On as-of-date lists the gain is +414%, not +1,076%. The 2026 test covers six months.

Check it yourself: the headline reproduces from scratch in about 6 minutes and prints ₹107,649,806. Every run, including failures, is in the repository.

With more time: a matched null band built from our own rule, inverse-volatility weighting, a full Indian cost model and regime sub-period tests

Source: docs/PROJECT.md §15; github.com/jashD11/finesseXcitadel. Team Finsearch · Arnab Tiwari, Mrunal Barapatre, Jash Dalal

Thank you

Questions?

Every number is reproducible: github.com/jashD11/finesseXcitadel

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